



Figure 2: Choosing packages during Onama installation

Mechanical engineering

Currently, the role of computers in mechanical engineering is very focused and limited. Numerous CAD/CAM software help mechanical engineers in the design, simulation, and analysis of products. For example, performing finite-element analysis on mechanical models is an indispensable process in almost all mechanical industries. Presently, Onama offers various applications to help mechanical engineers perform these tasks. Q-CAD and G-Simple cater to all basic CAD/CAM requirements. Gas Dynamics Calculator (GDC) is a utility used to calculate several quantities in different compressible flow models. After defining the necessary parameters, it provides calculations for isentropic and isothermal nozzles, Fanno and Rayleigh flows, and also shock and oblique shock—using direct calculations. To solve mechanical, transport, and fluid mechanics problems, engineers can use OOFEM, RHEOLEF, SLFFEA, Z88, Gerris and OpenFOAM. Mechanism lets you solve complex mechanical equations, and also plot the solutions in different forms.

Civil engineering

Civil engineering is a basic branch of engineering that doesn't rely heavily on computing. However, Onama contains a few applications to help civil engineers perform complex calculations. Aladdin and Frame3DD are very useful for the structural analysis of different types of cross-sections. JCAE is a Java-based environment for running different CAE applications. Euler, on the other hand, is powerful numerical software like Matlab. With the advent of more complex civil structures, it is expected that HPC will help civil engineers perform faster and more accurate analysis.

Electrical engineering

The electrical engineering component of Onama includes numerous software to calculate and predict electrical parameters. Basic circuit design can be accomplished using X-Circuit and Electric. Eq Resistor, Induction Motor and BPF help to calculate inductances, resistances, impedances,

frequencies, and resistor combinations for a designed circuit. Electronic Design is a tool-kit used for conversions between various electrical parameters. MICRO_VAPR is an application dedicated to the design of different types of inductors.

Electronics engineering

Logisim, PCB and Tk-Gate are useful applications for designing and simulating complex circuits as well as individual printed circuit boards. TranCalc is an analysis and synthesis tool that lets users calculate the electrical and physical properties of RF and microwave transmission lines. MEEP is a powerful finite-difference time-domain simulation software package used for modelling electromagnetic systems. It was developed by the Massachusetts Institute of Technology, and is very popular among electronics engineers. Dinotrace is a waveform viewer that understands numerous trace formats. It allows placing cursors, highlighting signals, searching, printing, and numerous other capabilities. For engineers using the NEC2antenna modelling software, Antennavis provides a simple and fast way to render NEC antenna outputs.

Chemical engineering

Most chemical engineers attribute their greatest developments to powerful computers that helped them perform inhuman calculations. Onama contains several molecular drawing programs to generate two- and three-dimensional representations. Avogadro, BkChem, REDMD, JMOL, and JChemPaint are designers and simulators for molecular dynamics. XmakeMOL is a simple mouse-based application useful for measuring bond lengths, bond angles, torsion angles, and such quantities. TOXTREE helps estimate the toxic effect of different chemical compounds. The Massively Parallel Quantum Chemistry (MPQC) program provides implementations for several important methods in quantum chemistry.

Computer engineering

Computer engineering and information technology revolve around the development of faster and better software for other sectors. Thus, this field has seen more open source development than any other. Onama includes the most popular computer engineering applications that are used by industry professionals.

Bluefish is a versatile Web development editor that supports numerous technologies. DDT is a database design creator that also allows the importing of SQL descriptions. BOUML is a very fast code generator for C++, Java, PHP, and Python. JUML is a reverse-engineering tool for decoding Java source code to create relationship diagrams based on UML. Neuroph is an object-oriented neural-network framework that can be used to create and train neural networks through Java. OS-Simulator is a versatile